

Decomposed B2 prediction and bounded LLM research triage

These are two distinct experiments. The first measures drug-disease ranking accuracy on TRANSCRIPT. The second asks an LLM to prioritize **next research actions** for the frozen LUAD Top-10. Its output is not a drug efficacy score and must not be merged with the NS-AUC table.

B2 component ablation

The frozen B2 model is an unweighted reciprocal-rank fusion of three inputs: B0p (training-fold positive count per drug), B1k (drug-expression kNN propagation of training-fold positive labels), and B1 (negative Spearman correlation between drug and disease expression signatures). The component adapters in `benchmarks/recess_adapter/official_components.py` exactly match the corresponding frozen TranscriptBaseline scores in unit tests; changing validation ratings leaves their scores unchanged. B0p and B1k use only training-fold associations.

The adapters enter the same pinned [RECESS official runner](#) as B2, using 100 identical seed values, a 20% outer holdout, five inner folds, and the author's NS-AUC calculation for both split types. The second patch is applied *after* `official_b2.patch` and does not modify B2's implementation. The published five-fold operation selects the best fold-trained model by inner row-wise AUC; it does not search a hyperparameter grid. The weakly correlated split repeats one outer holdout across seeds, so its spread is not uncertainty over 100 distinct outer splits.

To reproduce, apply both patches to a clean clone at commit `a7f11077271cedf3a98a82e3dc74b6fc0e93986e`, then run:

```
python -m scripts.run_recess_official_components --upstream $upstream --n 100 --k 5 --njobs 4
python -m scripts.compare_recess_components
```

The comparison checks all 100 seed positions, finite metrics, pairwise B2 differences, and source hashes. It writes `benchmark/results/recess_official_component_ablation.json`.

Frozen method	Random simple NS-AUC	Weakly correlated NS-AUC
B0p: training association count	0.5000 ± 0.0000	0.5000 ± 0.0000
B1k: expression-neighbor propagation	0.1747 ± 0.0393	0.0562 ± 0.0466
B1: label-free expression reversal	0.4816 ± 0.0340	0.5417 ± 0.0000
B2: fixed RRF fusion	0.5222 ± 0.0354	0.5019 ± 0.0036

Values are means ± sample SD over 100 upstream metric values. On random simple, B2 exceeds B1 by 0.0407 paired NS-AUC points and wins 97/100 seeds; on weakly correlated, B1 exceeds B2 by 0.0398 and wins 100/100 seeds. The weak-split B1 zero SD reflects the same deterministic outer

holdout, **not** evidence of negligible uncertainty. The B1k adapter was additionally checked against the frozen TranscriptBaseline on the real TRANSCRIPT first random split; all three component score vectors matched exactly (maximum absolute difference 0). The very low B1k NS-AUC is therefore not explained by an adapter score mismatch, though the dataset/metric cause has not been proven. The versioned aggregate CSVs and parameter files are in `benchmark/results/recess_official_components/`.

The B0p random run was resumed after changing only `njobs` from 1 to 4: the official runner reused the first 14 completed seed intermediates and computed the rest in parallel. Both parameter receipts are retained. Seed order, model code, folds and metric were unchanged; `njobs` affects execution scheduling only.

Before inspecting component ablation results, one accepted DeepSeek Flash call chose a method from the four frozen options using only the versioned method, split, and dataset descriptions in `configs/component_selector_v1.json`. It selected B2 for random simple and B1 for weakly correlated. The choices, input hash, timestamp and token usage were committed in `benchmark/results/recess_component_llm_selector_v1.json` at fbd9858 while the component runs were still in progress. An earlier strict-tool response failed the local reason-length check and was not saved; the accepted retry still preceded inspection of ablation scores. This is a single prescore choice, not a tuned classifier or a reliable estimate of method-selection skill. After the run, both choices rank first among these **four** methods: selected NS-AUC is 0.5222 for random simple and 0.5417 for weakly correlated. Relative to always using B2, the latter is +0.0398 on the same repeated outer holdout. There are only two split-level decisions, no candidate- or disease-specific LLM routing and no independent model-selection holdout. This result cannot support a claim that LLM selection generally improves prediction or beats the published 11-model panel. The model's random-split rationale described B0p/B1k label signal as reliable, but their observed NS-AUCs were 0.5000 and 0.1747. Thus even its successful method choice came with a partly unsupported explanation. The reasoning text must not be treated as an interpretation of the measured result.

LLM evidence triage

`scripts/run_luad_llm_adjudication.py` reads the frozen `configs/luad_top10_evidence_v1.json` and the human-reviewed `docs/luad_top10_evidence_matrix.md`. It does not read TRANSCRIPT labels or benchmark predictions. DeepSeek Flash is asked for one to three **research actions**, each tied to an existing rank and supplied PMID. A strict function schema and a local validator reject unknown drugs, fabricated/mis-scoped PMIDs, unknown actions, duplicate ranks, clinical actions, and an assay recommendation for a cross-source identity-unresolved compound. All ten evidence tiers remain `insufficient_evidence`; model prose is advisory and unverified.

The real call used the [DeepSeek strict tool-call mode](#) on 2026-09-26. Two earlier JSON-mode responses failed local validation and were not accepted or saved. The one accepted response prioritized rank 6 BRD-K84203638 and rank 5 flumetasone for identity resolution, and rank 2 beclomethasone-dipropionate for literature review. It did not propose a treatment or claim efficacy. The validated output and input hashes are saved in `benchmark/results/luad_llm_adjudication_`

v1.json. This is **one model sample**, not a blinded accuracy estimate or proof that its prioritization is optimal.

Run it again only with a local DEEPSEEK_API_KEY or an ignored .env file:

```
python -m scripts.run_luad_llm_adjudication
```

The key is never written to the result. Repeated model calls and blinded human review would be needed to quantify decision consistency and evidence quality.

Planner v3 is a separate metric

Agent v3's 98/100 is exact-match selection of the expected tool on a frozen 100-case set; it does not grade drug predictions. Its two failures are v3_luad_003 (package_luad_case expected, manual_review selected) and v3_boundary_004 (manual_review expected, rank_transcriptome selected). Both were valid tool calls, not API errors. The stored trace has no raw model reasoning, so their internal cause cannot be determined from this run. See docs/planner_eval_results.md.